

Third-Round Review Report

The True Cost of Volatility Targeting: Decomposing the Insurance Premium into Opportunity and Transaction Components

Document type: Journal paper (target: Journal of Portfolio Management / Financial Analysts Journal)

Review date: 2026-04-05

Review standard: Strict (focused re-review of Round 2 items)

Reviewer: Claude Opus 4.6 (1M context)

Previous reviews: review_v1.tex (Round 1), review_v2.tex (Round 2)

Review Summary

SEVERE issues (must fix)	1 item (unchanged: E3 sensitivity data)
MEDIUM issues (should fix)	2 items (down from 6 in Round 2)
Minor issues (optional)	3 items (down from 5)
Overall assessment	Good — near submission-ready

This revision makes substantial progress on the Round 2 issue list. Of the 6 medium items from Round 2, four have been resolved: W3/W10 (VoV literature: Bekaert & Hoerova 2014 and Todorov 2010 added), W7 (DM test footnote added with full specification), N2 (sub-period numbers now provided: $\rho = 0.04$, rebalancing premium 48 bps), and C2/C3/C4/C5 (citation corrections). The single remaining blocker is **E3**: the sensitivity analysis numbers in the paper (64%, 57%, 65% opp share; 62–76% total reduction) still cannot be traced to any recorded JSON file. The user indicates that three separate per-threshold JSON files (`k811v2_th0_5/1_0/1_5_results.json`) were intended but **do not yet exist** in the experiments directory.

Part A: Round 2 Priority Issue Resolution

E3: Sensitivity numbers — STILL NOT FIXED (SEVERE)

Round 2 complaint: Paper claims opp shares of 64%, 57%, 65% at thresholds 0.5, 1.0, 1.5. Existing JSONs (`k811v2_sensitivity_results.json`, `k811v2_sensitivity_sweep.json`) show opp shares of 81–89% and total reductions of –8% to –31%.

What was done: The paper text is unchanged at line 196. The user states that three separate JSON files (`k811v2_th0_5_results.json`, `k811v2_th1_0_results.json`, `k811v2_th1_5_results.json`) were intended to contain the per-threshold results using the *main-analysis decomposition methodology*. However, these files **do not exist** in `experiments/`.

Analysis of the discrepancy: The confusion arises from two different definitions of “opp share”:

- **Main-analysis definition** (used in Table 2): $\text{OppShare}(S_i) = \text{IP}_{\text{opp}}^{S_i} / \text{IP}_{\text{total}}^{S_i}$. For S2 at threshold 1.0: $0.696 / 1.218 = 57.1\%$. This is what the paper reports.
- **Sweep-JSON definition:** appears to compute something different—the sweep JSON reports `s2_opp_share = 85.0%` at threshold 1.0, which cannot be $0.696 / 1.218$.

The paper’s 57% for threshold 1.0 *is consistent* with the main results JSON (`k811v2_insurance_premium_vo`) which reports S2 opp = 0.696%/yr, total = 1.218%/yr. The 64% (threshold 0.5) and 65% (threshold

1.5) presumably come from re-running the identical decomposition at those thresholds—but those runs were never saved.

Required action:

1. Run the main K811v2 decomposition code with `vov_threshold` set to 0.5, 1.0, and 1.5 separately.
2. Save each run to its own JSON file (e.g., `k811v2_th0_5_results.json`, etc.).
3. Verify the paper’s numbers (64%, 57%, 65% opp share; 62–76% total reduction) match the JSON outputs.
4. If they match: the blocker is resolved. If they don’t: update the paper.

Conditional assessment: If the 64/57/65% numbers are confirmed by proper JSON files using the main-analysis definition, then E3 is resolved and the paper becomes submission-ready. The main-analysis definition is the correct one for the paper’s narrative (“what fraction of S2’s own premium is opportunity cost?”). The sweep JSONs appear to use a different calculation and are not relevant to the paper’s claims.

Verdict: **SEVERE until JSONs are created and verified.** Likely resolvable in < 1 hour.

Part B: Round 2 MEDIUM Issue Resolution

#	Issue	Status	Comment
M1	E1 residual: 5-day window & S3 blend 0.5	Not fixed	Still no justification for the 5-day VIX-rising window or the 0.5 blending coefficient. These are minor for practitioner journals but would be flagged in academic review.
M2	W7: DM test details	FIXED	A footnote now appears at line 193 with full specification: “DM tests use QLIKE loss on squared returns as the loss function, with Newey-West HAC standard errors (bandwidth = $\lfloor 4(T/100)^{2/9} \rfloor$). We report two-sided t -statistics and apply the Harvey (2016) multiple-testing threshold of $ t > 3.0$.” This is complete and correct.
M3	C2: Huang reference	FIXED	Line 263 now reads: “Huang, D., Schlag, C., Shaliastovich, I., Thimme, J., 2019. Volatility-of-volatility risk. <i>JFQA</i> 54, 2423–2452.” All 4 authors, correct journal, correct year. The bibkey remains <code>huang2015</code> internally but displays as “Huang et al. (2019)” via <code>\bibitem[Huang et al.(2019)]{huang2015}</code> , which is cosmetically acceptable.

M4	C5: Cederburg characterization	FIXED	Line 56 now reads: “cederburg2020 document poor out-of-sample performance across 103 volatility-managed strategies.” This accurately describes their primary finding (OOS failure) rather than the previous overstatement about transaction costs.
M5	N2: Sub-period claim	FIXED	The footnote at line 183 now provides concrete numbers: “The 2012–2024 sub-period yields $\rho = 0.04$ and a rebalancing premium of 48 bps, broadly consistent with the full-sample estimates.” This replaces the unsubstantiated “both estimates are stable across sub-periods” assertion.
M6	W3/W10: VoV literature	FIXED	Two key references added to the bibliography: Bekaert & Hoerova (2014, <i>Journal of Econometrics</i> 183, 181–190) and Todorov (2010, <i>Review of Financial Studies</i> 23, 345–383). Both are cited at line 205 alongside the existing Bollerslev et al. (2009) and Huang et al. (2019). The VoV literature section is now adequately covered for a practitioner journal.

Summary: 5 of 6 Round 2 medium issues resolved. Only M1 (5-day window and S3 blend justification) remains, and it is minor.

Part C: Citation Check Resolution

#	Issue	Status	Comment
C1	Bongaerts content claim	Fixed (R1)	“conditioning on volatility states” — correct since Round 1 revision.
C2	Huang 2015 WP → 2019 JFQA	FIXED	Now shows 4 authors, 2019, JFQA 54, 2423–2452 (line 263). Internal bibkey still huang2015 but display text correct.
C3	Perchet pages 21–36 → 21–38	FIXED	Line 273 now reads “21–38.”
C4	Liu “smooth sailing” quotes	FIXED	Line 205 now reads “smooth sailing environments” without LaTeX quotation marks (``...'' removed). The phrase is now clearly the authors’ paraphrase rather than a direct quotation.
C5	Cederburg characterization	FIXED	See M4 above.

All 5 citation-check items from Round 2 are now resolved.

Part D: Full Re-Check of Key Claims Against Data

D1: Main decomposition numbers (Table 2)

Strategy	Paper (Table 2)		JSON (insurance_cost_summary)	
	Opp (%/yr)	Direct (%/yr)	Opp (%/yr)	Direct (%/yr)
S1	4.20	0.43	4.195	0.428
S2	0.70	0.52	0.696	0.522
S3	2.85	0.46	2.854	0.456

All match within rounding (paper rounds to 2 decimal places).

D2: Performance metrics (Table 1)

Spot-checked against full_period_metrics in the main results JSON:

	Paper		JSON	
	CAGR	Sharpe	CAGR	Sharpe
S0 (BH)	12.51	0.60	12.506	0.5984
S1	7.11	0.54	7.111	0.5353
S2	11.14	0.63	11.144	0.6335
S3	8.84	0.57	8.842	0.5699
S4	7.89	0.50	7.888	0.5013

All match.

D3: DM test statistics

Paper (line 193) reports: “S1 vs. S0: $t = 2.42$; S2 vs. S0: $t = 0.75$.”

JSON (dm_tests): S1 vs S0 $t = 2.42$ (matches); S2 vs S0 $t = 0.75$ (matches). **Match.**

D4: Cross-OOS

Paper: “S2 outperforms BH SPY in only 1 of 4 periods.” JSON: s2_wins_s0: 1. **Match.**

D5: Sensitivity analysis (E3 — THE BLOCKER)

Paper (line 196): opp shares 64%, 57%, 65%; total reduction 62–76%.

Threshold	Paper Opp Share	sensitivity_results	sweep (s2_opp_share)	Main JSON (S2)
0.5	64%	81.7%	81.4%	—
1.0	57%	85.5%	85.0%	57.1%
1.5	65%	89.1%	89.1%	—

The paper’s 57% at threshold 1.0 *matches* the main-analysis decomposition ($0.696/1.218 = 57.1\%$), confirming the paper uses the “share of S2’s own premium” definition. The sweep/sensitivity JSONs use a *different* definition (likely share of S1-equivalent premium at that threshold). The 64% and 65% values are not recorded in any existing JSON.

For the total reduction:

- The paper implies S2 at threshold 0.5 reduces total cost by 62% (and at 1.5 by 76%) relative to S1.
- The sweep JSON shows `total_reduction = -30.5%` (threshold 0.5), `-13.1%` (1.0), `-8.5%` (1.5) — these are in the *wrong direction* (S2 costs *more* at each threshold in the sweep), confirming the sweep uses a different methodology.
- The main analysis shows S2 total = 1.218%/yr vs S1 total = 4.623%/yr, giving a reduction of $1 - 1.218/4.623 = 73.7\%$, matching the paper’s “74%” claim. But reductions at thresholds 0.5 and 1.5 are not recorded.

Verdict: The paper’s numbers are plausible and use a consistent definition, but the threshold-0.5 and threshold-1.5 runs have not been saved. Three new JSON files are needed to close this item.

Part E: New Reference Verification

Two new references were added. Verification:

Bekaert and Hoerova (2014) — bekaert2014

Field	Paper	Verified
Authors	Bekaert, G., Hoerova, M.	Geert Bekaert, Marie Hoerova ✓
Year	2014	2014 ✓
Title	The VIX, the variance premium and stock market volatility	Correct ✓
Journal	<i>Journal of Econometrics</i>	Journal of Econometrics ✓
Volume	183	183(2) ✓
Pages	181–190	181–190 ✓

Content claim: Cited at line 205 as part of the VoV risk literature. Bekaert & Hoerova decompose VIX^2 into expected variance and variance risk premium, directly relevant to VoV discussion. **PASS.**

Todorov (2010) — todorov2010

Field	Paper	Verified
Authors	Todorov, V.	Viktor Todorov ✓
Year	2010	2010 ✓
Title	Variance risk-premium dynamics: The role of jumps	Correct ✓
Journal	<i>Review of Financial Studies</i>	Review of Financial Studies ✓
Volume	23	23(1) ✓
Pages	345–383	345–383 ✓

Content claim: Cited at line 205 as VoV risk literature. Todorov studies variance risk premium dynamics with jumps—directly relevant. **PASS.**

Part F: Remaining Issues (Consolidated)

SEVERE (must fix before submission)

#	Location	Description
E3	Sec 3.5, line 196	Sensitivity analysis numbers lack JSON backing. Paper claims opp shares of 64%, 57%, 65% and total reduction 62–76%. Only the 57% (threshold 1.0) can be traced to the main results JSON. The 64% (threshold 0.5) and 65% (threshold 1.5) have no recorded source. Create three per-threshold JSON files by re-running the main decomposition code. If the numbers match: resolved. If not: update paper.

MEDIUM (should fix)

#	Location	Description
M1	Sec 2.2	S3 blend coefficient 0.5: Still presented without justification. One sentence acknowledging this as a heuristic choice (“we select 0.5 as a simple midpoint; sensitivity to this parameter is left for future work”) would suffice.
M7	Sec 2.2	5-day VIX-rising window: No explanation of why 5 days rather than 10 or 20. Add a brief note (e.g., “we use a 5-day window to capture short-term momentum in VIX, corresponding to one trading week”).

Minor (optional)

#	Location	Description
L1	Eq (3)/ Sec 2.3	N symbol conflict: Eq. (3) uses N for years, but text uses $N = 3,262$ for trading days. Use T for years.
L6	References	Bibkey huang2015 displays correctly as “Huang et al. (2019)” but the internal key is misleading. Consider renaming to huang2019 for maintenance clarity. Not visible to readers.
L5	Throughout	“apples-to-apples” appears twice (Sec 3.3 and Sec 4). Mildly colloquial for academic prose; consider replacing one instance with “directly comparable.”

Progress Summary Across Three Rounds

Category	Round 1	Round 2	Round 3
SEVERE	5	1	1
MEDIUM	8	6	2
Minor	7	5	3
Total	20	12	6

Issues resolved this round: W3/W10 (VoV literature), W7 (DM footnote), N2 (sub-period numbers), C2 (Huang → 2019), C3 (Perchet pages), C4 (smooth sailing quotes), C5 (Cederburg characterization).

Issues resolved in previous rounds: E1 (partially), E2, E4, E5, W1, W4, W5, W8, W9, C1.

Still open: E3 (sensitivity JSONs), M1 (S3 blend), M7 (5-day window), L1/L5/L6 (minor notation/style).

Overall Assessment

This paper is substantively strong and nearly submission-ready. The core contribution—91% of VT’s insurance premium is opportunity cost, not transaction cost—is novel, clearly stated, and fully verified against the main experiment JSON. The VoV conditioning finding is appropriately hedged as preliminary. The literature is now adequate with the addition of Bekaert & Hoerova (2014) and Todorov (2010). All citation errors have been corrected. The DM test specification is now complete.

The single blocker is E3: three sensitivity runs at thresholds 0.5, 1.0, and 1.5 need to be saved to JSON files so the paper’s sensitivity claims are traceable. The paper’s numbers (64%, 57%, 65%) are *plausible*—the 57% matches the main analysis exactly, and the definition is consistent—but “plausible” is not “verified.” Once the JSONs exist and confirm, the paper can be submitted.

Estimated effort to reach submission-ready:

1. **E3:** Re-run threshold sweep with main decomposition, save 3 JSONs, verify. **~30 minutes.**
2. **M1 + M7:** Add one sentence each for S3 blend and 5-day window justification. **~10 minutes.**
3. **Minor (L1/L5/L6):** Notation and bibkey cleanup. **~10 minutes.**

Target journal recommendation: **Journal of Portfolio Management** (primary) or **Financial Analysts Journal** (secondary). The practitioner-oriented framing, clear decomposition, and actionable implications are well-suited to these venues. For JFE/RFS, additional work would be needed: formal hypothesis testing, walk-forward validation, and a more extensive theoretical framework connecting VoV conditioning to the variance risk premium literature.